

γ -hadron correlations for $Pb + Pb$ collisions at $\sqrt{s_{NN}} = 5.02$ TeV

• Setting for the numerical calculations

TABLE I. Configurations for the NLO pQCD calculations.

system	$Pb + Pb$ collisions
colliding energy	$\sqrt{s_{NN}} = 5.02$ TeV
centrality	0-30%, 30-50%, 50-90%
rapidity	$ y^\gamma < 0.67$, $ y^h < 0.9$
away-side range	$ \Delta\phi > 3\pi/5$
$p_T^{\text{trig}}(\gamma)$	[18-40] [GeV/ c]
$p_T^{\text{assoc}}(h^\pm)$	[> 0.5] [GeV/ c]
z_T	[0.15-0.2], [0.2-0.3], [0.3-0.4], [0.4-0.6], [0.6-0.8], [0.8-1.0]
PDFs	CT18A PDF [1]
FFs	Kniehl-Kramer-Potter FFs[2]
energy loss	High-twist approach [3] with $\hat{q}/T^3(T)$ extracted from single hadron, dihadron and γ -hadron data by the IF-Bayesian analysis[4, 5]

• Numerical results

In $p + p$ collisions, the triggered fragmentation functions $D_{pp}^{\text{trig}}(z_T)$ is defined as [6],

$$D_{pp}^{\text{trig}}(z_T) = p_T^{\text{trig}} \frac{\frac{d\sigma_{pp}^{\gamma h}}{dy^{\text{trig}} dp_T^{\text{trig}} dy^{\text{assoc}} dp_T^{\text{assoc}}}}{\frac{d\sigma_{pp}^{\gamma}}{dy^{\text{trig}} dp_T^{\text{trig}}}}. \quad (1)$$

where the numerator is the cross section of triggered-hadron production, the denominator is the cross section of single inclusive trigger particle cross section. In $A + B$ collisions,

$$D_{AB}^{\text{trig}}(z_T) = p_T^{\text{trig}} \frac{\frac{d\sigma_{AB}^{\gamma h}}{dy^{\text{trig}} dp_T^{\text{trig}} dy^{\text{assoc}} dp_T^{\text{assoc}}}}{\frac{d\sigma_{AB}^{\gamma}}{dy^{\text{trig}} dp_T^{\text{trig}}}}. \quad (2)$$

The nuclear modification factors of γ -hadron are defined as,

$$I_{AB}(z_T) = \frac{D_{AB}(z_T)}{D_{pp}(z_T)}. \quad (3)$$

In the ALICE presentation of QM 2023, we have noticed some discrepancies between our pQCD results and the ALICE data. Such discrepancies may primarily originate from the following two aspects.

Firstly, according to Eq. (2), the cold nuclear matter (CNM) effects in both photon and hadron-photon yields should ideally cancel out. However, in our calculations, the cross-section for photon-hadron productions as a function of the x distribution ($d\sigma^{\gamma h}/dx$) corresponds to a typical x value that is slightly larger than those in the cross-section for photons ($d\sigma^{\gamma}/dx$). This precisely corresponds to the anti-shadowing effect region exhibited by photon-hadron correlations being slightly stronger than photons, leading to a nuclear modification factor slightly greater than 1 when only considering the CNM effects. This issue was also discussed in previous emails. To avoid this influence, our recalculated results this time have included only the energy loss effects, similar to the approach taken by the CoLBT model.

Secondly, the results we previously provided for 0-30% were with a simple average hydro of 0-10%, 10-20%, and 20-30% centrality bins, and for 50-90% were with the average hydro of 50-70% and 70-90%. To better reflect a more realistic collision scenario, we have now weighted the nuclear modification factors I_{AA} for each centrality by the corresponding γ -hadron cross-sections to obtain the averaged results. For I_{AA} of 0-30% centrality,

$$I_{AA}(0-30\%) = \frac{(I_{AA} * d\sigma^{\gamma h})(0-10\%) + (I_{AA} * d\sigma^{\gamma h})(10-20\%) + (I_{AA} * d\sigma^{\gamma h})(20-30\%) }{d\sigma^{\gamma h}(0-10\%) + d\sigma^{\gamma h}(10-20\%) + d\sigma^{\gamma h}(20-30\%)},$$

and

$$I_{AA}(50-90\%) = \frac{(I_{AA} * d\sigma^{\gamma h})(50-70\%) + (I_{AA} * d\sigma^{\gamma h})(70-90\%) }{d\sigma^{\gamma h}(50-70\%) + d\sigma^{\gamma h}(70-90\%) }.$$

Below are the results after implementing these two adjustments.

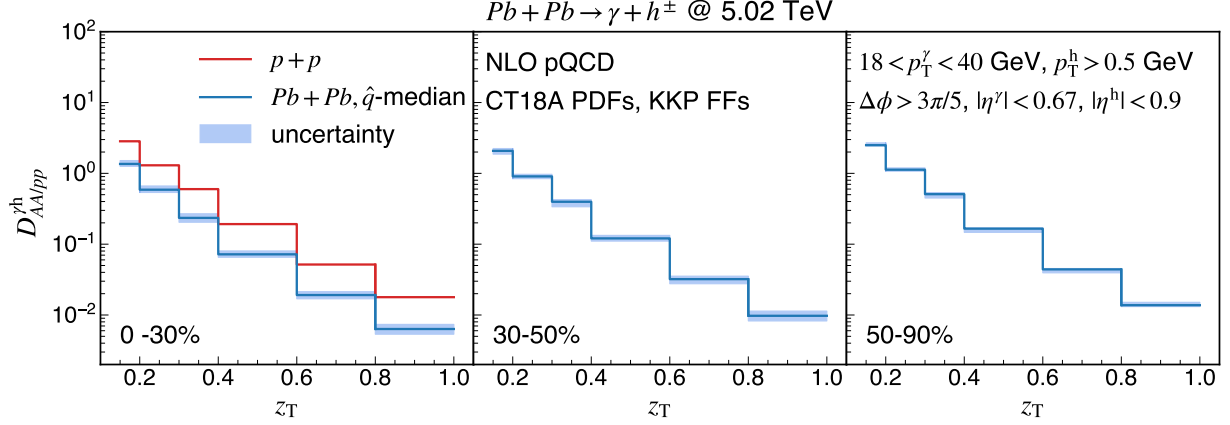


FIG. 1. γ -triggered fragmentation functions in $p + p$ collisions and $Pb + Pb$ collisions with 0-30%, 30-50%, and 50-90% centralities at $\sqrt{s_{NN}} = 5.02$ TeV. The red curve is the NLO pQCD result for $p + p$ collisions. The blue curves are the results with the median value of $\hat{q}/T^3$ extracted from the single hadron, dihadron and γ -hadron data with the IF-Bayesian inference method [4, 5], and the light blue bands are the estimated uncertainty.

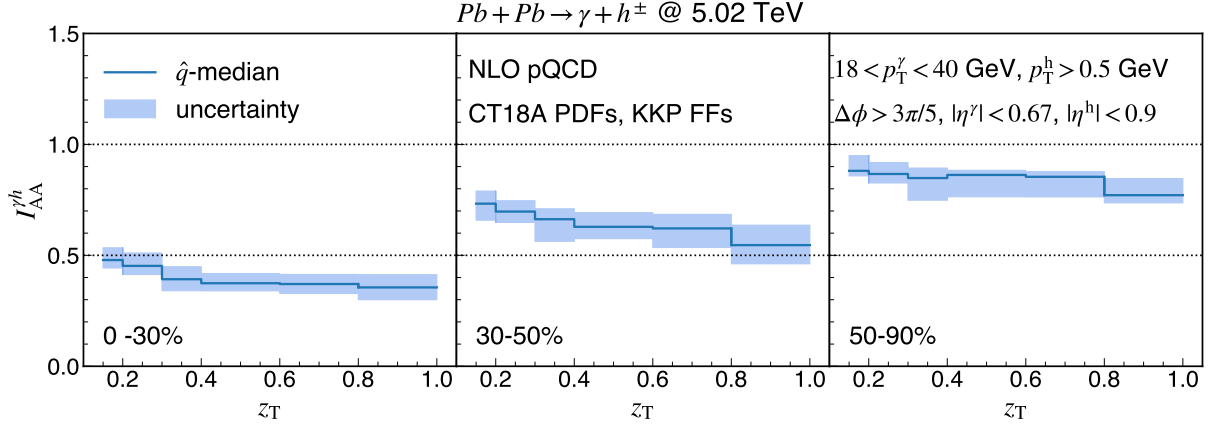


FIG. 2. γ -hadron suppression factors I_{AA} as a function of z_T in $Pb + Pb$ collisions at $\sqrt{s_{NN}} = 5.02$ TeV with 0-30%, 30-50%, and 50-90% centralities, respectively. The blue curves are the NLO pQCD results with the median value of $\hat{q}/T^3$ extracted from the single hadron, dihadron and γ -hadron data with the IF-Bayesian inference method [4, 5], and the light blue bands are the estimated uncertainty.

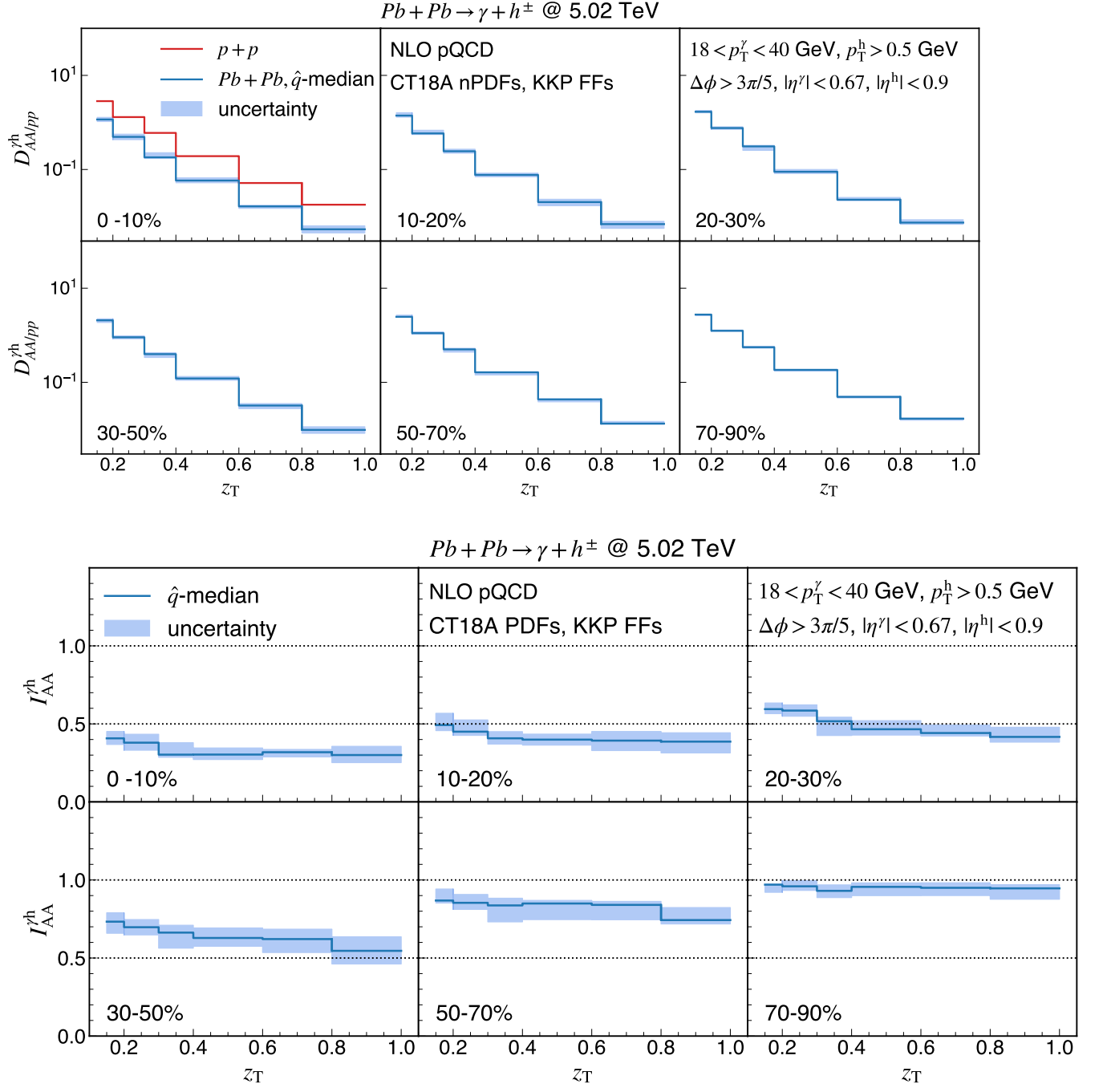


FIG. 3. The corresponding results for 0-10%, 10-20%, 20-30%, 30-50%, 50-70%, and 70-90% centrality bins.

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